Chapter 12

5-Day Practice Set

Radicals, Pythagorean Theorem & Square Root Functions

Name: _____ **Date:** _____ **Period:** _____

Day	Topics	Score	Date
Day 1	Foundation	/10	
Day 2	Developing	/10	
Day 3	Intermediate	/10	
Day 4	Advanced	/10	
Day 5	Challenge	/10	

Topics covered each day:

- **A.** Finding the missing side of a right triangle (Pythagorean Theorem)
- **B.** State whether the given lengths can be sides of a right triangle
- **C.** Simplify radical expressions
- **D.** Solve radical equations
- **E.** Graph square root functions

Day 1 Foundation

10 problems · Easy

Section A: Find the Missing Side Length

Use $a^2 + b^2 = c^2$, where c is the hypotenuse.

1. $a = 3,\ b = 4.$ Find c.

2. $a = 5,\ c = 13.$ Find b.

Section B: Is It a Right Triangle?

Check whether $a^2 + b^2 = c^2$ holds (use the longest side as c).

3. $8,\ 15,\ 17$

4. $3,\ 5,\ 6$

Section C: Simplify Each Radical Expression

5. $\sqrt{28}$

6. $\sqrt{75}$

7. $\sqrt{2} + \sqrt{8}$

Section D: Solve the Radical Equation

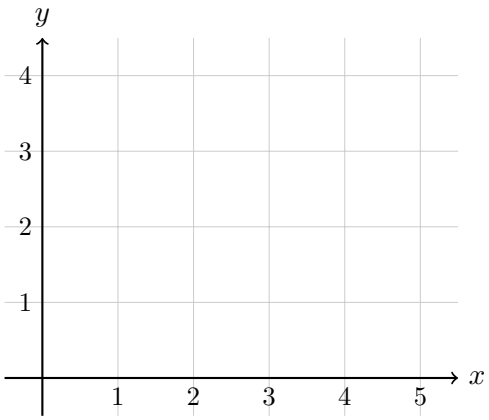
8. $\sqrt{x} = 6$

9. $\sqrt{x} - 1 = 4$

Section E: Graph the Square Root Function

10. Graph $y = \sqrt{x}$. Complete the table, then sketch.

x	0	1	4	9
y				



Domain: _____ Range: _____

Day 2 Developing

10 problems · Easy–Medium

Section A: Find the Missing Side Length

1. $a = 9,\ b = 12.$ Find c.

2. $b = 7,\ c = 25.$ Find a.

Section B: Is It a Right Triangle?

3. $1.5,\ 2.0,\ 2.5$

4. $5,\ 6,\ 8$

Section C: Simplify Each Radical Expression

5. $\sqrt{300}$

6. $3\sqrt{5} + \sqrt{45}$

7. $4\sqrt{10} - \sqrt{40}$

Section D: Solve the Radical Equation

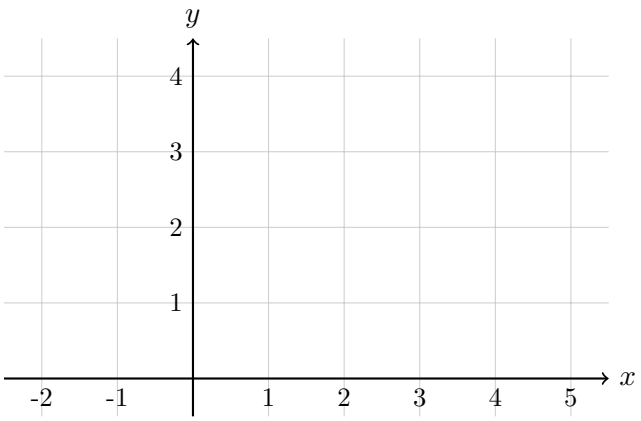
8. $\sqrt{x+3} = 5$

9. $2\sqrt{x} = 10$

Section E: Graph the Square Root Function

10. Graph $y = \sqrt{x+2}$. Complete the table, then sketch.

x	-2	-1	2	7
y				



Domain: _____ Range: _____

Day 3 Intermediate

10 problems · Medium

Section A: Find the Missing Side Length (Round to Nearest Tenth)

1. $a = 7,\ b = 9.$ Find c.

2. $a = 4.0,\ c = 4.1.$ Find b.

Section B: Is It a Right Triangle?

3. 7, 24, 25

4. 8, 16, 17

Section C: Simplify Each Radical Expression

5. $\sqrt{3} + \sqrt{12}$

6. $\dfrac{-\sqrt{18}}{\sqrt{12}}$

7. $\dfrac{1}{\sqrt{3}+4}$ (rationalize the denominator)

Section D: Solve the Radical Equation. Check your solution.

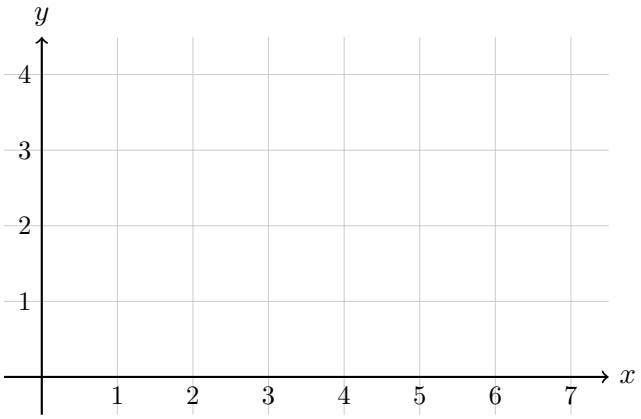
8. $\sqrt{2x}+3=7$

9. $\dfrac{\sqrt{3m+2}}{3}=1$

Section E: Graph the Square Root Function

10. Graph $y=\sqrt{x-3}$. Complete the table, then sketch.

x	3	4	7	12
y				



Domain: _____ Range: _____

Day 4 Advanced

10 problems · Medium–Hard

Section A: Find the Missing Side Length

1. In a right triangle, one leg = 20 and the hypotenuse = 21. Find the other leg. Round to the nearest tenth.

2. **Word problem:** A 17-foot ladder leans against a wall. The base of the ladder is 8 feet from the wall. How high up the wall does the ladder reach?

Section B: Is It a Right Triangle?

3. 0.9, 1.2, 1.5

4. 11, 13, 17

Section C: Simplify Each Radical Expression

5. $\dfrac{\sqrt{6}}{4-\sqrt{6}}$ (rationalize the denominator)

6. $\dfrac{\sqrt{2}}{\sqrt{2}+3}$ (rationalize the denominator)

7. $-3\sqrt{5x^3} \cdot \sqrt{10x^3}$

Section D: Solve the Radical Equation. Check for extraneous solutions.

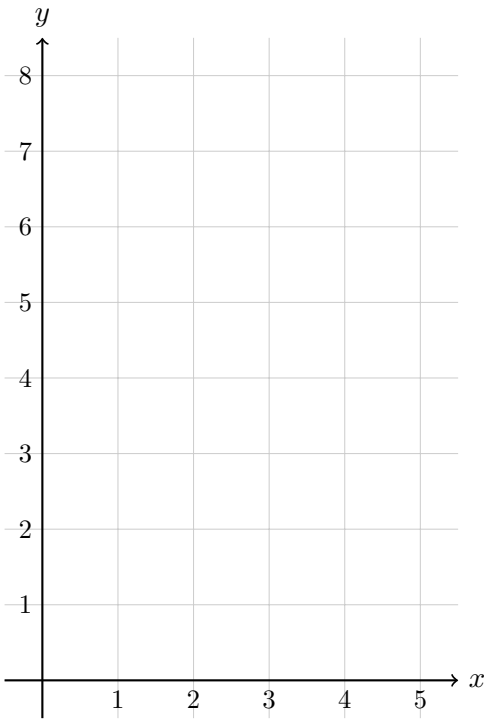
8. $\sqrt{2x+4} = \sqrt{3x}$

9. $\sqrt{2-x} = x$

Section E: Graph the Square Root Function

10. Graph $y = \sqrt{x} + 5$. Complete the table, then sketch.

x	0	1	4	9
y				



Domain: _____ Range: _____ Starting point: _____

Day 5 Challenge

10 problems · Hard

Section A: Find the Missing Side Length

1. **Word problem:** A baseball diamond is a square with 90-foot sides. How far is it from home plate to second base? (Hint: the diagonal of the square is the hypotenuse.) Round to the nearest tenth.

2. **Word problem:** Two hikers leave from the same point. One walks 12 miles north and the other walks 16 miles east. How far apart are they?

Section B: Is It a Right Triangle? (These involve non-integer sides.)

3. $\sqrt{2},\ \sqrt{3},\ \sqrt{5}$

4. $\sqrt{3},\ 2,\ \sqrt{6}$ (Is $(\sqrt{3})^2 + 2^2 = (\sqrt{6})^2$?)

Section C: Simplify Each Radical Expression

5. $\dfrac{\sqrt{5} - \sqrt{3}}{\sqrt{5} + \sqrt{3}}$ (rationalize the denominator)

6. $\dfrac{3}{2\sqrt{3}-\sqrt{2}}$ (rationalize the denominator)

7. $\sqrt{2x+4}\cdot\sqrt{8x+16}$ (simplify completely)

Section D: Solve the Radical Equation. Check for extraneous solutions.

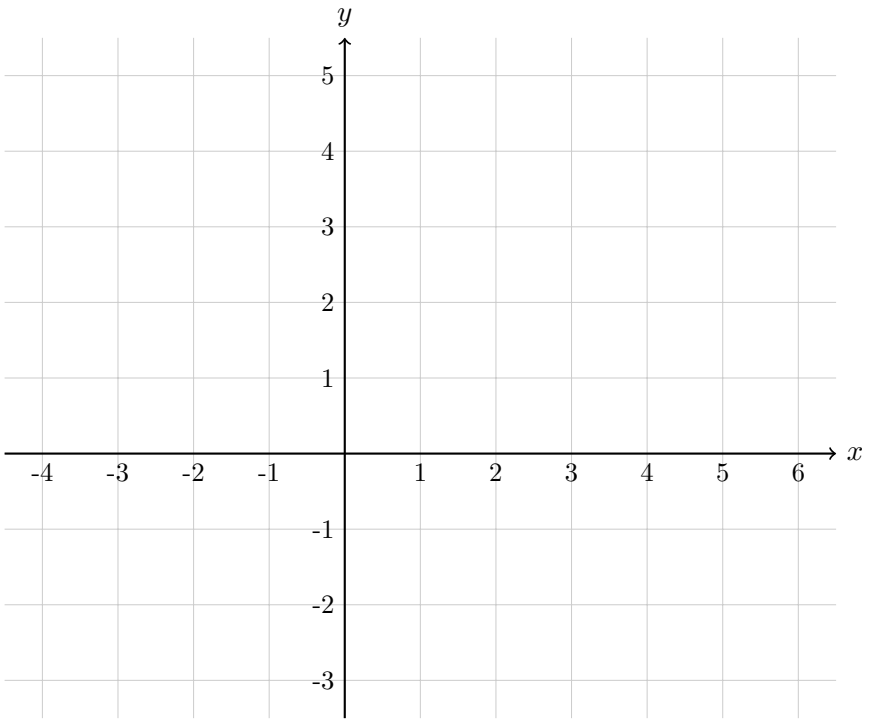
8. $\sqrt{-5a+6}=-a$

9. $\sqrt{x+5}+\sqrt{x}=5$

Section E: Graph the Square Root Function

10. Graph $y=\sqrt{x+4}-2$. Complete the table, then sketch.

x	-4	-3	0	5	12
y					



Domain: _____ Range: _____ Starting point: _____

Answer Key

Day 1: 1. $c = 5$ 2. $b = 12$ 3. Yes 4. No 5. $2\sqrt{7}$ 6. $5\sqrt{3}$ 7. $3\sqrt{2}$ 8. $x = 36$ 9. $x = 25$ 10. Domain $x \geq 0$, Range $y \geq 0$

Day 2: 1. $c = 15$ 2. $a = 24$ 3. Yes 4. No 5. $10\sqrt{3}$ 6. $6\sqrt{5}$ 7. $2\sqrt{10}$ 8. $x = 22$ 9. $x = 25$ 10. Domain $x \geq -2$, Range $y \geq 0$

Day 3: 1. $c \approx 11.4$ 2. $b = 0.9$ 3. Yes 4. No 5. $3\sqrt{3}$ 6. $-\dfrac{\sqrt{6}}{2}$ 7. $\dfrac{4-\sqrt{3}}{13}$ 8. $x = 8$ 9. $m = \dfrac{7}{3}$ 10. Domain $x \geq 3$, Range $y \geq 0$

Day 4: 1. ≈ 6.4 2. 15 ft 3. Yes 4. No 5. $\dfrac{2\sqrt{6}+3}{5}$ 6. $\dfrac{3\sqrt{2}-2}{7}$ 7. $-15x^3\sqrt{2}$ 8. $x = 4$ 9. $x = 1$ (check: $x = -2$ is extraneous) 10. Domain $x \geq 0$, Range $y \geq 5$, Start $(0, 5)$

Day 5: 1. $90\sqrt{2} \approx 127.3$ ft 2. 20 miles 3. Yes $(2 + 3 = 5)$ 4. No $(3 + 4 = 7 \neq 6)$ 5. $4 - \sqrt{15}$ 6. $\dfrac{6\sqrt{3}+3\sqrt{2}}{10}$ 7. $4|x+2|$ 8. $a = -6$ ($a = 1$ extraneous) 9. $x = 4$ 10. Domain $x \geq -4$, Range $y \geq -2$, Start $(-4, -2)$